

SURESH KOSANA

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PROFESSIONAL SUMMARY

Fresher Software Engineer (MCA, 2025) with strong depth in Java, Data Structures & Algorithms, and backend development using Spring Boot, Spring Security, and REST APIs. Built three backend projects spanning JWT-secured REST APIs with 97% test coverage and a distributed object-synchronization system implementing CRDTs and vector clocks from scratch. Actively strengthening DSA through daily practice; eager to contribute to a backend engineering team.

TECHNICAL SKILLS

Languages: Java (Core/SE), SQL

Frameworks: Spring Boot, Spring Security, Spring Core, Spring Data JPA, Hibernate, Maven

Web & APIs: REST APIs, JWT Authentication, JSON, Swagger/OpenAPI, Postman, HTML5, CSS3

Databases: MySQL, PostgreSQL — joins, queries, schema design

Distributed Systems: CRDTs, Vector Clocks, TCP Socket Programming, Multithreading, Concurrency

Testing: JUnit 5, Mockito, JaCoCo (test coverage)

Data Structures: Arrays, Strings, Linked Lists, Stacks, Queues, Searching & Sorting

Tools: Git, GitHub, Docker, Eclipse, VS Code, Spring Tool Suite 4 (STS), pgAdmin4, MySQL Workbench 8.0 CE, ModelMapper, Bean Validation

PROJECTS

Student Management System 2026

Spring Boot 3.5, Java 21, Spring Security, JWT, Spring Data JPA, MySQL, JUnit 5, Docker

- Built a role-based Student Management REST API using Spring Boot, Spring Security, and JWT authentication with stateless session management, enforcing USER/ADMIN access control across endpoints.
- Designed a clean DTO layer decoupling API contracts from JPA entities across all CRUD, search, sort, and pagination endpoints, using ModelMapper for entity-DTO conversion.
- Implemented centralized exception handling (@RestControllerAdvice) with granular custom exceptions (invalid credentials → 401, resource not found → 404, validation errors → 400) for consistent API responses.
- Wrote 77 unit, controller, and integration tests (JUnit 5, Mockito) achieving 97% instruction coverage verified via JaCoCo; documented the API with Swagger/OpenAPI and containerized the application with Docker.

Quiver: Consistent Object Sharing System 2025

Java, Distributed Systems, CRDTs, Vector Clocks, TCP Sockets, Multithreading, Maven, JUnit 5

- Designed and implemented a 3-node distributed object-sharing system in Java using vector clocks for causality tracking and a Last-Write-Wins CRDT for automatic, conflict-free convergence across replicas.
- Built a multi-threaded TCP socket networking layer with JSON-based (Gson) peer-to-peer messaging and a gossip-style sync protocol, verified across three live node processes.
- Fixed a concurrency bug by migrating the shared object store to ConcurrentHashMap, since it was accessed simultaneously by the CLI thread and per-connection network threads.
- Wrote 16 JUnit 5 tests covering CRDT convergence, causal ordering, and conflict tie-breaking, replacing earlier ad hoc manual verification.

Placement Management System 2024

Spring Boot, PostgreSQL, Spring Data JPA/Hibernate, Bean Validation

- Built a RESTful Placement Management API using Spring Boot and Spring Data JPA with PostgreSQL, supporting full CRUD operations on student placement records.
- Designed a DTO layer to decouple API contracts from JPA entities, with Bean Validation (@NotBlank, @NotNull, @Positive) enforcing data integrity, and centralized exception handling returning consistent 400/404 responses.

EDUCATION

Master of Computer Applications (MCA) — CGPA: 7.32/10 2023 – 2025

Adikavi Nannaya University (S.V.K.P & Dr. K.S. Raju Arts & Science College), Penugonda

B.Sc. Computer Science — CGPA: 7.18/10 2020 – 2023

Adikavi Nannaya University (S.V.K.P & Dr. K.S. Raju Arts & Science College), Penugonda

PROFESSIONAL TRAINING

Java Full Stack Developer Training — TNS India Foundation, Chennai Oct 2024 – Mar 2025

- Hands-on training in full-stack development using Java and Spring Boot; built backend logic and database integrations across multiple mini-projects, practicing debugging and collaborative development with Git and GitHub.

CERTIFICATIONS

Full Stack Java Development — Capgemini & TNS India Foundation (Cert ID: T030305224) • AI Fundamentals, Project Management Fundamentals, Job Application Essentials — IBM SkillsBuild